Assignment 3: Monitor Unencrypted S3 Buckets Using AWS Lambda and Boto3

Objective

The objective of this assignment is to enhance AWS security by identifying Amazon S3 buckets that do not have explicit server-side encryption configurations enabled. This is implemented using AWS Lambda and the Boto3 Python SDK.

AWS Services Used

Amazon S3

AWS Lambda

IAM

Boto3 (Python SDK)

Amazon CloudWatch

Implementation Steps

1. Created an Amazon S3 bucket named unencrypted-bucket-rohan for encryption audit testing.
2. Created an IAM role with AmazonS3ReadOnlyAccess permission for Lambda execution.
3. Created a Lambda function using Python 3.12 runtime.
4. Implemented Boto3 logic to list all S3 buckets and check server-side encryption status.
5. Manually triggered the Lambda function to validate the encryption status of buckets.

Lambda Function Logic

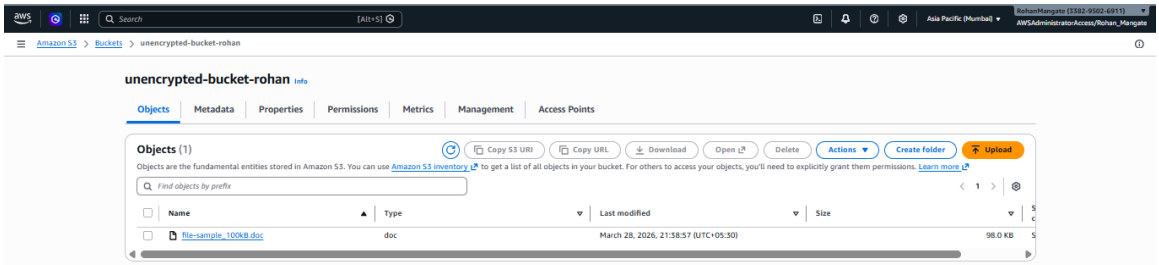
The Lambda function uses the boto3 S3 client to list all buckets within the AWS account. For each bucket, the function attempts to retrieve its encryption configuration using the get_bucket_encryption API. Buckets that do not return an encryption configuration are flagged and logged for review. Execution logs are written to Amazon CloudWatch.

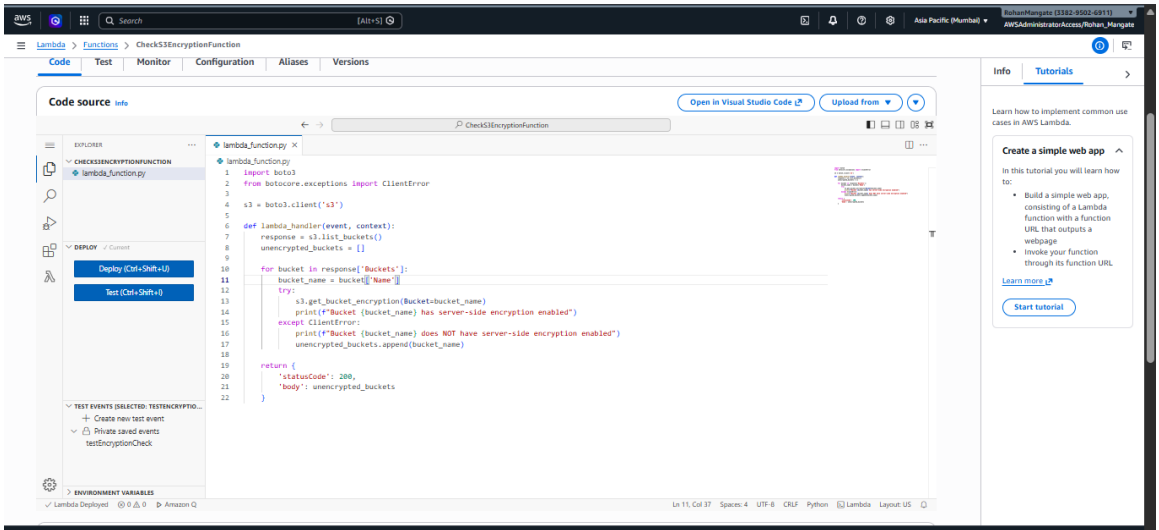
Testing and Validation

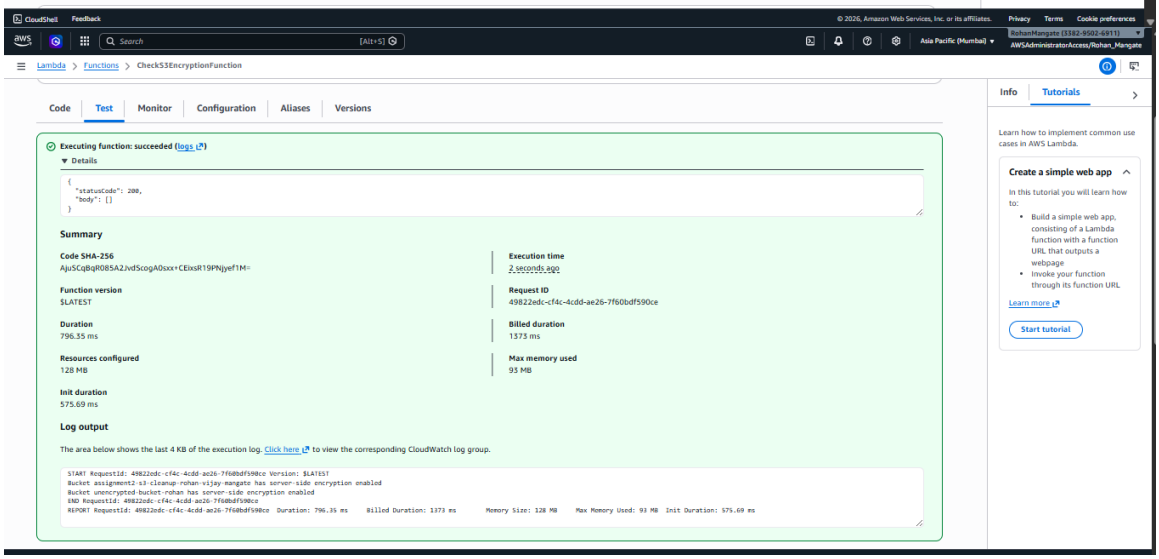
The Lambda function was manually invoked using a test event. The execution completed successfully and logged the encryption status of each S3 bucket. Due to AWS default settings, new buckets are automatically encrypted with SSE-S3, which was verified through logs.

Screenshots

The following screenshots are included as part of the assignment submission:
1. S3 bucket creation screen showing unencrypted-bucket-rohan.
2. Lambda function Python code for encryption check.
3. Lambda test execution result showing successful run.
4. CloudWatch logs displaying bucket encryption status.







Conclusion

This assignment successfully demonstrates how AWS Lambda and Boto3 can be used to audit Amazon S3 bucket encryption settings. The solution meets the security monitoring objective by programmatically validating server-side encryption and logging results for review.